When we need to extend / interconnect LAN, we can use:-

1.> Repeater:-

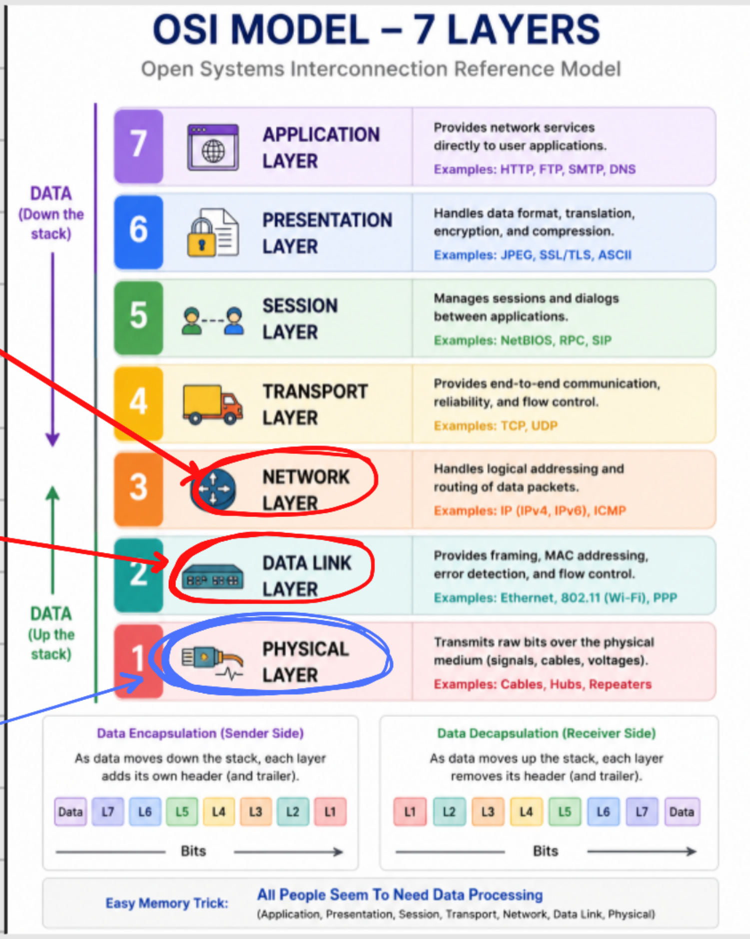
→ Connect identical LANs
↑ using same MAC protocols.

Repeaters don't process frames, they operate at physical layer because they operate at physical layer, they only regenerate Raw Signal.

Router works at this Layer

Bridges work at this layer.

Repeater Works at this layer



2.> Bridges :-

Interconnect LANs that uses similar or different MAC Protocols.

→ Bridges do process frames

→ Works at Data Link Layer

→ Works at Data Link Layer.

3) Routers :-

→ Similar to bridges + extra functionalities.

→ Works at Network Layer.

Repeaters :-

→ it does not connect two LAN's.

→ Overcome the distance Limitation of LANs.

→ Example an Ethernet Standard 10base5 mean a repeater is needed beyond 500m.

10 Mbps   baseband   500m

→ Regenerate attenuated bits into fresh signals.

Bridges

→ it connects small LANs to create one large LAN.

→ Advantages of Using Small LANs.

- Reliability : Faults are Contained within a small segment.
- Performance : Small LANs improve speed by keeping all traffic Local.
- Security : Bridges isolate traffic so not all stations can see all data.
- Geography : Allows LAN to extend across distant location.

Function OF A Bridge

Basic Bridge

→ only understands one frame format.
→ They are sometimes called MAC Relay Bridges.

It provides following functionalities:-

1) Store And Forward

→ operates in promiscuous mode i.e reads all incoming frames even the ones not meant for it.

→ It forwards a frame to another LAN if the destination device is located on that LAN.

→ This function is performed in both directions.

LAN A ⇄ Bridge ⇄ LAN B

→ Frames are copied bit-by-bit.

2) Routing And Addressing

→ Frames relevant to a particular LAN Segment are copied

So.... storing a frame upon arrival is a part of Store-and-Forward.
and decides where to retransmit it using Routing-and-addressing.

* Devices don't change the way they communicate just because a bridge is present.

Routing Strategy is used to decide which frames are forwarded onto another LAN.

1) Fixed Routing:-

Bridge has a pre-made table like this:-

Destination MAC	which LAN it belongs
B	LAN 2
C	LAN 1

→ Table is Manually set.

Advantages / Disadvantages

Adv

Simplicity: Requires Minimum processing overhead.

DisAdv

Maintenance Hell: Requires manual intervention when more station/bridges are added or Removed.

2) Address Learning Routing Strategy

→ Bridge learns the location of each station automatically.

→ Each incoming MAC frame contain a source address field.

→ Each LAN attaches to one port only.

Steps

→ initially routing table is empty

→ MAC frame arrives →

bridge checks routing table for destination address

Address found in table

↓

Source MAC / port recorded

↓

frame sent to correct port

Dest Not found

↓

Source MAC / port recorded

↓

frame broadcasted to all other ports.

→ After a period, "Steady-State period," the table is complete.

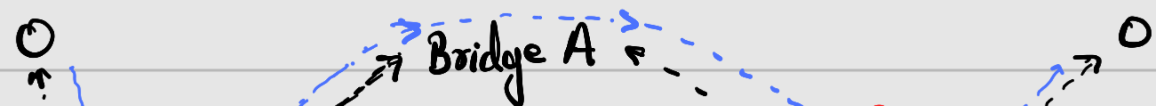
→ Now <u>Frame-Filtering</u>, takes place

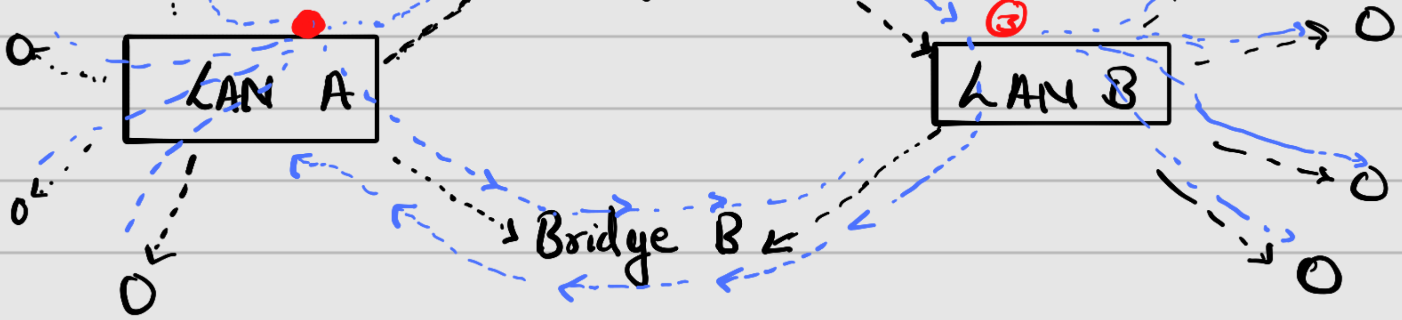
↑ bridge stops forwarding unnecessary frames

Bridges that use address learning strategy are called "Transparent Bridges"

Problems with transparent bridges:-

<u>Looping</u>





Frame → LANA → Bridge A → LAN B → Bridge B

Overcoming Bridging Loops

↳ achieved by creating a spanning tree

≃

Advantages of Bridges

1> Parallelism:-

A bridge doesn't forward every frame, so different LAN can communicate independently at the same time.

device in LAN A
can talk to another device
in LAN in parallel to a device on LAN B talking to another device
on LAN B

2) Optimized Performance

→ Stations that communicate with each other often can be moved to one LAN.

→ This ensures : Principle of Locality of Reference.

→ Collision Domain are smaller, as hosts are now localised into sub LANS.

* Without bridging :-

Each station perceives the speed of LAN as:-

= LAN Data Rate / (numbers of stations on the whole network)

With bridging :-

Each station perceives the speed of LAN as:-

= LAN Data Rate / number of stations on that smaller LAN